Oxford College of Emory University

Chapter 14, Part 1

Money, Banks, and the Federal Reserve

Guilherme Paiva Pinto

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Deposits and Withdrawals

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

What happens if many depositors request their money at the same time?

Roadmap

1. Money, liquidity, and monetary measures
2. Banks and balance sheets
3. Bank runs and safeguards
4. The Fed and nonbank finance

Money, Income, and Wealth

Concept	Meaning	Example
Money	Assets readily usable for payment	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 of assets − $2,000 of debts

Money and Liquidity

Item	What is it?	Liquidity
Checking deposit	Money available for payment	Readily spendable
Debit card or check	Instruction accessing a deposit	Not an additional money balance
House	A real asset included in wealth	Selling takes time and can involve costs

The Functions of Money

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- **Unit of account**: states prices, wages, and debts in a common unit.
- **Store of value**: carries purchasing power into the future; inflation can erode it.

Types of Money

Type	Defining feature	Example
Commodity money	The payment item also has other uses	Gold, silver, or cigarettes
Commodity-backed money	A claim redeemable for a specified commodity	A paper claim redeemable for gold
Fiat currency	No fixed promise of commodity redemption	U.S. dollar notes

These arrangements have coexisted; this is not a universal historical timetable.

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- A payment app can move existing balances. Count the underlying money once.

U.S. Money Measures

Measure	Main components
M1	Public currency, demand deposits, and other liquid deposits, including savings deposits
M2	M1 plus small time deposits and retail money market mutual fund balances

Savings deposits have been included in M1 since May 2020.

Source: Federal Reserve Board, H.6 definitions.

What Banks Do

- Accept deposits and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

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- Accept deposits and provide payment services.
- Make loans and hold securities, connecting finance to spending and investment.
- Manage a **liquidity mismatch**: deposits can leave before long-term assets are repaid.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Liabilities: loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

A Bank Balance Sheet

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Reserves and the Monetary Base

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- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
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A customer deposits $200 of currency held outside banks.

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Reserves and deposits each rise by $200.

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Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

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A New Bank Loan

- The bank approves a $120 loan and credits the borrower’s deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank meet payments now?	Assets cannot all be sold immediately at little cost.
Solvency	Do asset values cover liabilities?	Losses can exhaust equity.

Forced asset sales can turn liquidity pressure into losses and weaker confidence.

Silicon Valley Bank: 2022–2023

- **Concentrated funding**: technology-sector customers held many large, uninsured deposits.

Source: Federal Reserve Board, SVB review, April 2023.

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- **Concentrated funding**: technology-sector customers held many large, uninsured deposits.
- **2022**: higher interest rates reduced the value of longer-term securities as deposits began leaving.
- **March 2023**: announced losses and a capital plan were followed by a rapid run and closure on March 10.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Runs and Deposit Insurance

- Withdrawals can exhaust liquid funds; hurried asset sales may create losses.

Source: FDIC, Deposit Insurance at a Glance.

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- The **Federal Deposit Insurance Corporation (FDIC)** insures eligible deposits, reducing covered depositors' incentive to withdraw first.

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Bank Runs and Deposit Insurance

- Withdrawals can exhaust liquid funds; hurried asset sales may create losses.
- The **Federal Deposit Insurance Corporation (FDIC)** insures eligible deposits, reducing covered depositors' incentive to withdraw first.
- FDIC coverage is generally $250,000 per depositor, per insured bank, per ownership category.

Source: FDIC, Deposit Insurance at a Glance.

Moral Hazard and Bank Capital

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- **Capital requirements**: banks must maintain a loss-absorbing equity cushion.
- **Supervision**: checks risk management alongside liquidity and capital.

The Federal Reserve System

Institution	Role
Board of Governors	Federal oversight and system responsibilities
12 Reserve Banks	Regional operations, financial services, and supervision
Federal Open Market Committee (FOMC)	Decisions about the monetary-policy stance

1913: the Federal Reserve Act established the Fed to address banking-system stresses.

Source: Federal Reserve Board, The Fed Explained.

Federal Reserve Tools

Tool	Main purpose
Discount window	Lending secured by pledged assets supports eligible institutions' liquidity.
Interest on reserve balances (IORB)	Interest paid on Fed reserve balances helps guide short-term rates.

Liquidity lending is available beyond crises; it does not erase loan losses.

Source: Federal Reserve Board, Discount Window.

Source: Federal Reserve Board, Interest on Reserve Balances FAQ.

Shadow Banking

- Some nonbank institutions connect borrowers with investors and other lenders.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

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Shadow Banking

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- Short-term funding of less liquid assets can create run-like vulnerabilities.
- Risk depends on assets, funding, and borrowing relative to equity; nonbank activities differ.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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Reserves +$100; deposits +$100.

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Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Money and Bank Claims

- A deposit is money to its holder and a liability of the bank.

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- Thursday: follow money creation through lending, settlement, repayment, and the deposit model.

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Chapter 14, Part 2

Money Creation and Bank Balance Sheets

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Roadmap

1. Loan creation and payment settlement
2. Repayment, withdrawals, and losses
3. A conditional deposit-expansion model
4. Practice and an individual checkpoint

Bank Accounting

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

Reserves and loans are assets. Deposits are liabilities. Equity is the residual.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
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Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

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The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Reserves remain 200; equity remains 100.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

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Bank A: reserves −120; deposits −120.

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Bank A: reserves −120; deposits −120.

Bank B: reserves +120; seller’s deposits +120.

The Banking System

Change since before the loan	Bank A	Bank B	Combined
Reserves	−120	+120	0
Loans	+120	0	+120
Deposits	0	+120	+120

The payment moves deposits; the loan created the additional $120 of deposits.

Cash Withdrawals

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- Public currency rises by $50, offsetting the fall in deposits.
- Currency + deposits is unchanged; the bank has less liquidity.

Principal Repayment

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- Loans fall by $50 and deposits fall by $50.
- Currency + deposits falls by $50; reserves and equity do not change.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
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The bank writes off $80 of unrecoverable loans.

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Loans fall to 820; equity falls to 20.

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Loans fall to 820; equity falls to 20.

Deposits and reserves remain unchanged at the write-off.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.

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- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.
- **Risk and profitability**: borrowers must qualify, and lending must cover costs.

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A Reserve-Ratio Model

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- All loan proceeds are redeposited; no extra reserves or currency leakage.
- Banks lend each available amount; borrowers want the loans; other constraints do not bind.

Successive Deposit Rounds

Round	New deposit ($)	Reserves retained	New loan
1	100.00	10.00	90.00
2	90.00	9.00	81.00
3	81.00	8.10	72.90

The central bank first buys a $100 security from a nonbank seller, adding reserves and a deposit.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.

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- **Additional loans**: \$900. Total extra reserves: \$100.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.
- These are limiting model results under the stated conditions.

Depositing Existing Currency

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Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

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With the same expansion conditions, deposits rise by 1,000 in total, but currency + deposits rises by 900.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.

Source: Federal Reserve Board, Reserve Requirements.

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- Banks still need liquidity and capital and must manage funding and credit risk.

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The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.
- $1/r$ at $r = 0$ does not predict infinite lending or an infinite money supply.

Source: Federal Reserve Board, Reserve Requirements.

Central-Bank Security Purchases

Transaction	Reserves	Nonbank deposits
Buy from a bank	Bank reserves rise	No immediate required change
Buy from a nonbank seller	Seller's bank reserves rise	Seller's deposit rises

Both transactions exchange assets; the immediate effect depends on the seller.

Balance-Sheet Practice

Use the worksheet's bank: reserves 150, loans 750, securities 100, deposits 900, equity 100.

Record a new $90 loan, then its payment to another bank.

For each event, verify assets = liabilities + equity and state what happens to total deposits.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

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Loans −30 and equity −30.

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Deposits and reserves do not change solely because of the write-off.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.

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- The multiplier depends on its model conditions.
- Chapter 15 connects central-bank policy, interest rates, spending, and AD.